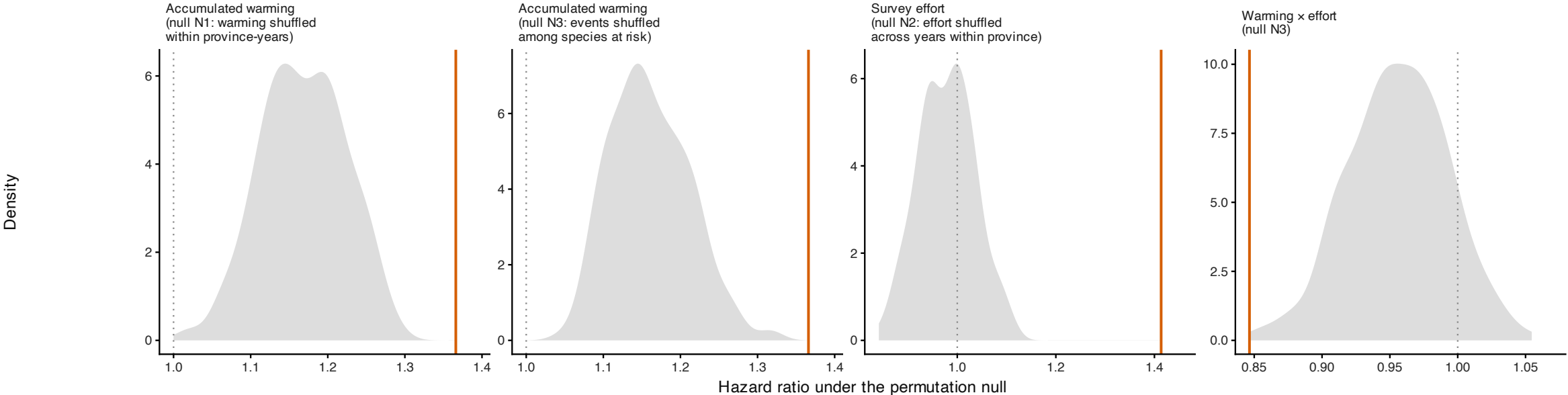


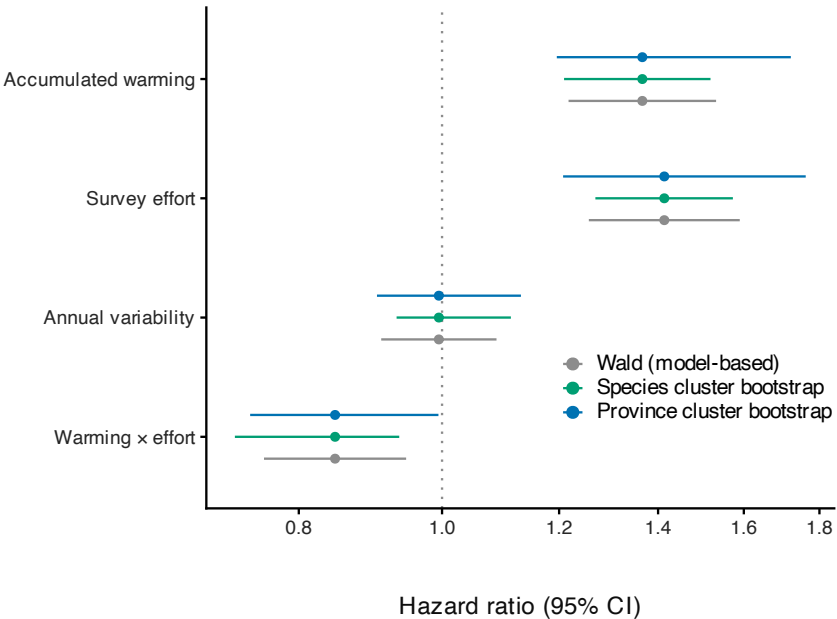
a Design-preserving permutation nulls (199 each)

Red: observed. Dotted: unity. The warming nulls centre near 1.16, not unity, because within-stratum shuffles preserve the provincial-annual climate–event covariance



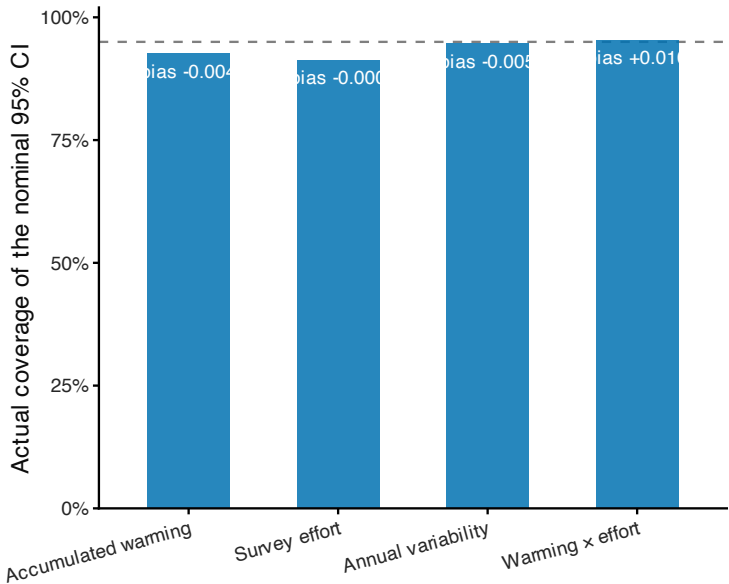
b Cluster bootstraps (300 resamples each)

Provincial clustering widens the intervals (SE ratio 1.26–1.59); species clustering does not (SE ratio 1.05–1.15)



c Parametric bootstrap (150 simulations)

Observed bias is not statistically significant



d Influence

No single province or species decile carries the result; maximum change 4.2%, 9 of 164 beyond 3% (dotted)

